

Physical Chemistry (II)

Lecture 22

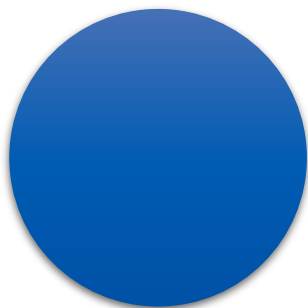
Polyatomic Molecules (1)



Last class

Electronic States of Molecules

- Determines the energy levels
- Described by molecular term symbols



Atom

Spherical symmetry

$\vec{L}_{\text{total}}^2$ conserved

L is a good quantum number



Diatomic molecule

Cylindrical symmetry

$L_{\text{total},z}$ conserved

M_L is a good quantum number

Molecular Term Symbol

$$^{2S+1}[|M_L|]$$

- Symbols for $[|M_L|]$

$ M_L =$	0	1	2	3
	Σ	Π	Δ	Φ

- $M_L = m_{l1} + m_{l2} + \dots, M_S = m_{s1} + m_{s2} + \dots$
- Completely filled orbitals do not contribute

Symmetry of Term Symbol

- g/u : sign change upon inversion

- can be obtained from molecular orbitals

$$g \cdot g = g, \quad g \cdot u = u, \quad u \cdot u = g$$

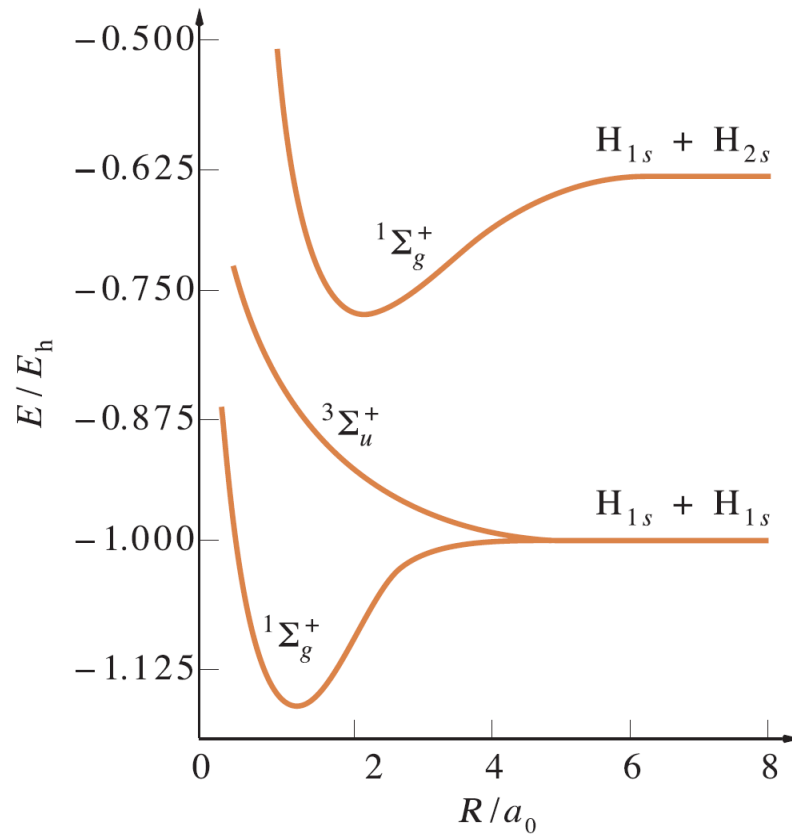
- $+/-$: sign change upon reflection through an internuclear plane (only applies to Σ electronic states!)

- can be obtained from molecular orbitals

$$(+)(+) = (+), \quad (+)(-) = (-), \quad (-)(-) = (+)$$

Last class

Potential Energy Curves

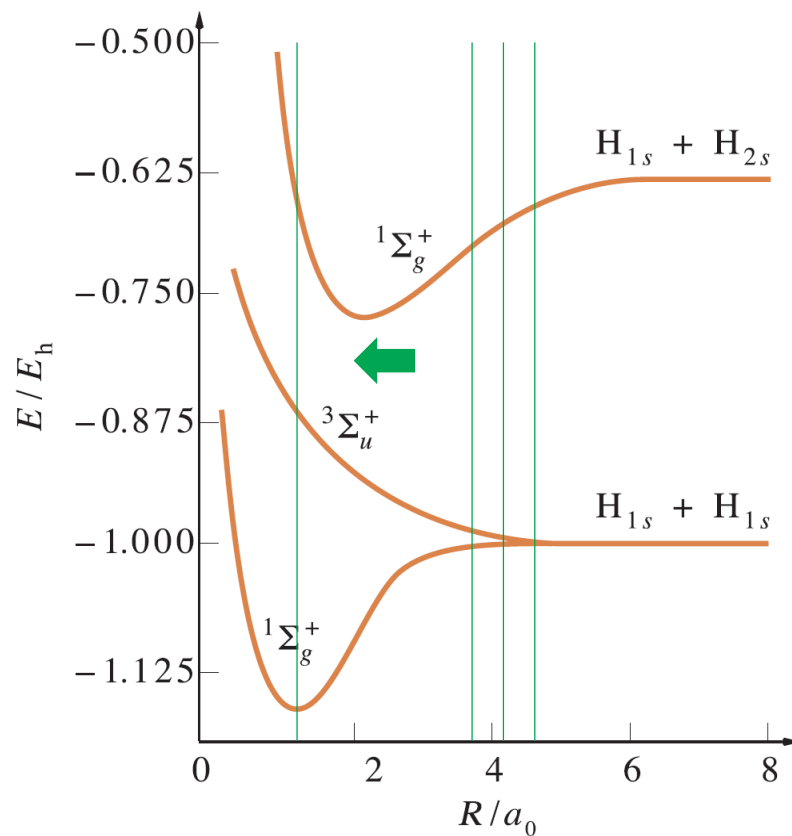


(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)



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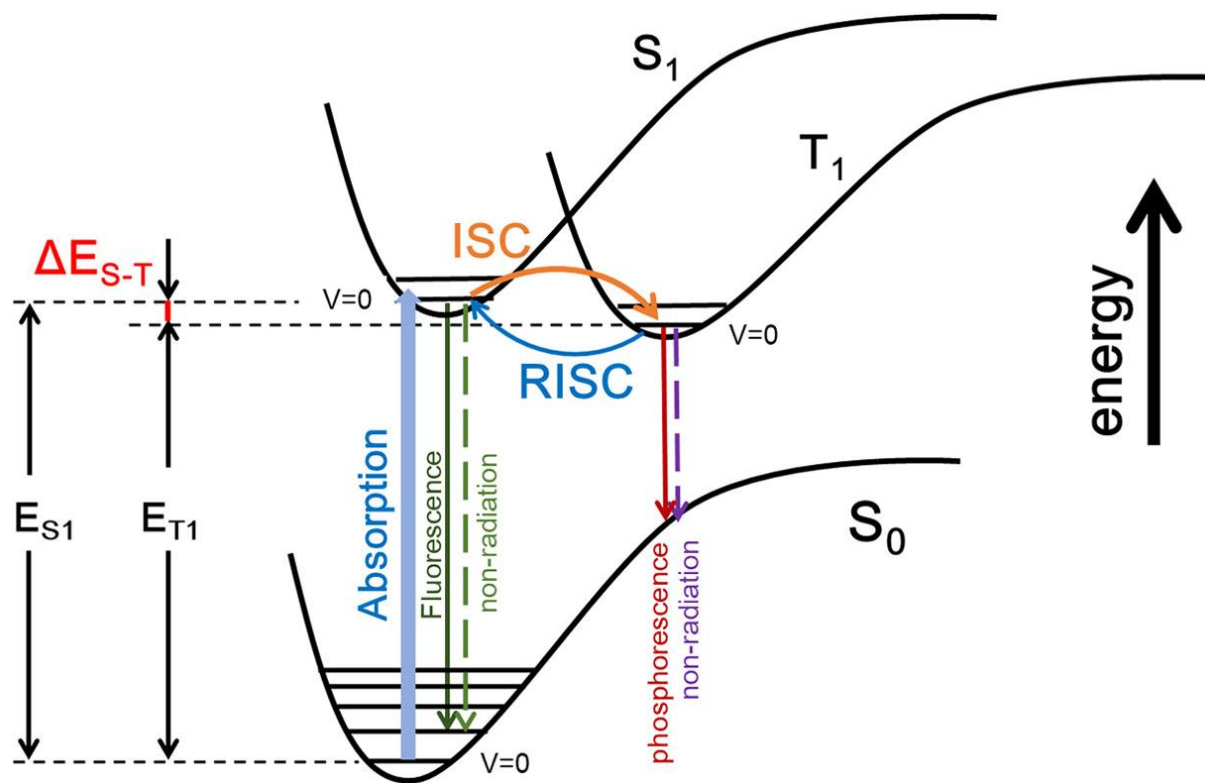
Geometry Optimization



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

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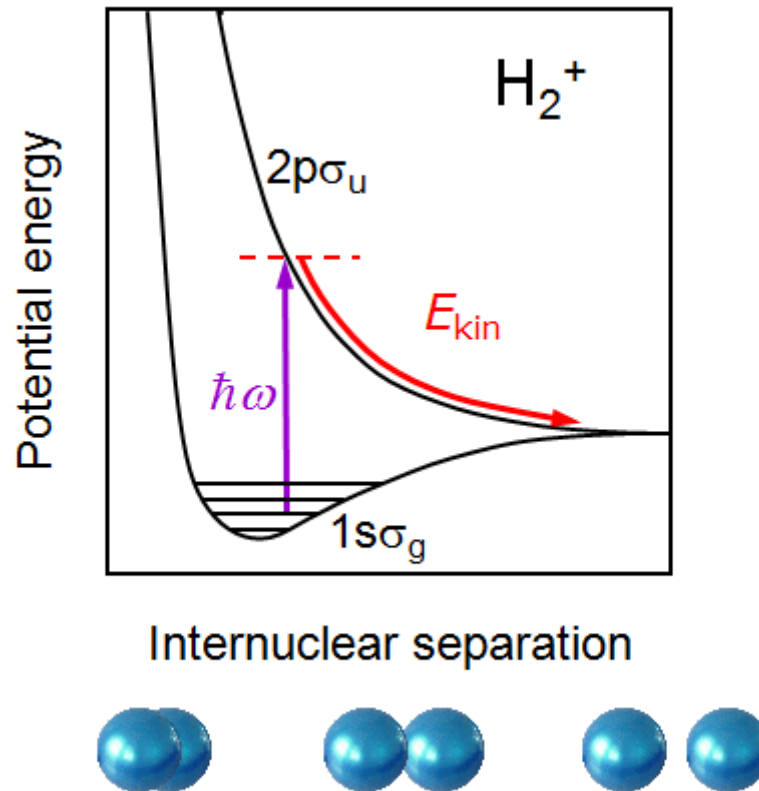
Fluorescence and Phosphorescence



(Image: Zheng et al., *J. Phys. Chem. Lett.*, 2022)

Last class

Photodissociation



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Construction Rules for MO

In order to construct MOs from two basis orbitals,

1. their overlap integral must be large enough

$$S_{ij} = \int \psi_i^* \psi_j d\tau > 0$$

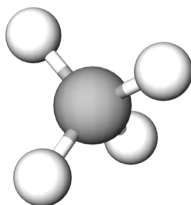
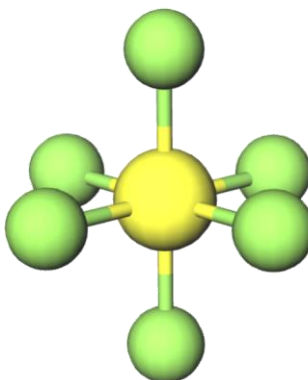
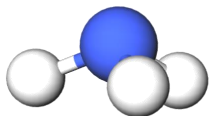
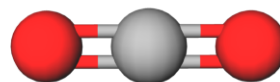
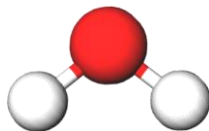
2. The energy levels must be close enough

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\int \psi_m^{(0)*} \hat{H}' \psi_n^{(0)} d\tau}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

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Main Targets

center atom + outer atoms

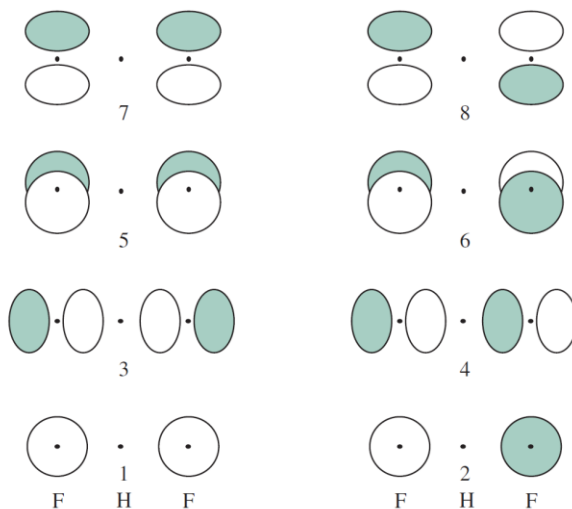


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Ingredients to Construct MOs

- Center atom: AOs
- Outer atoms: combinations of AOs = group orbitals

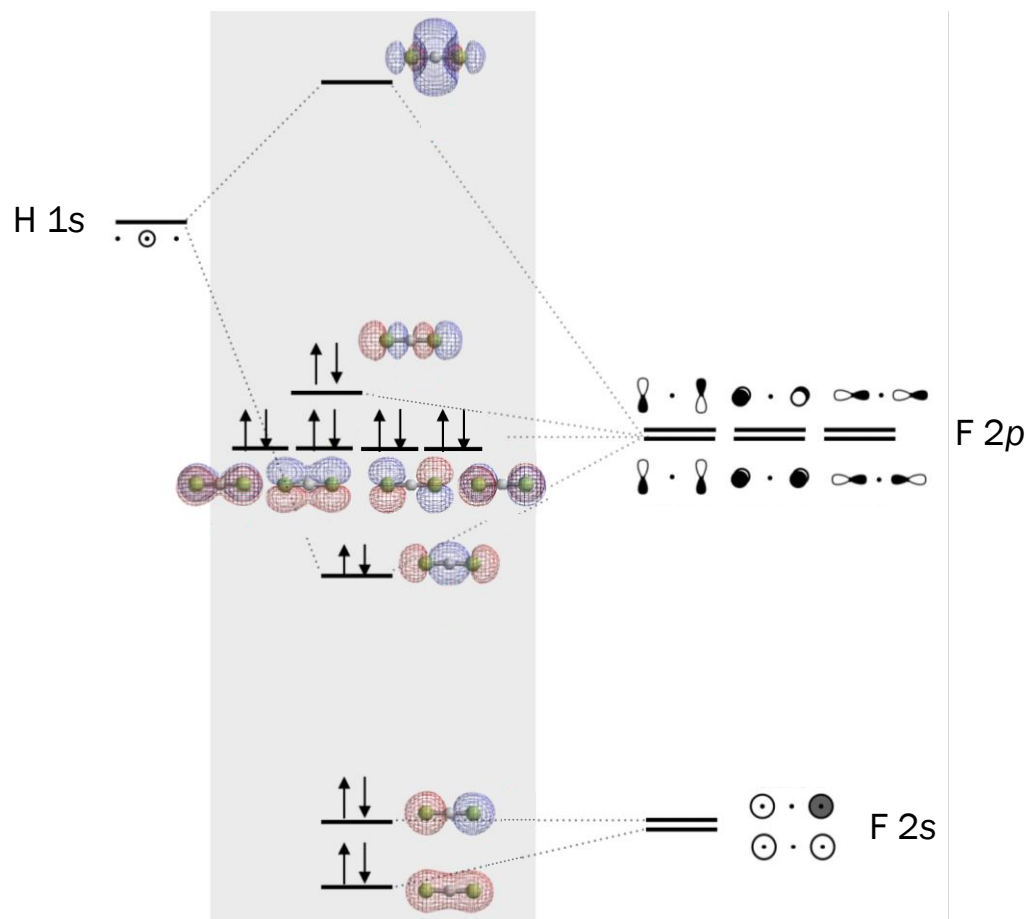
e.g. FHF^- (bifluoride anion)



(Image: Miessler et al., *Inorganic Chemistry*, 5th Ed., Figure 5.15)

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MO Diagram of FHF⁻

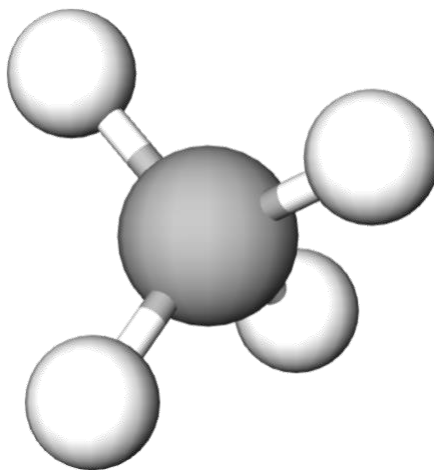


(Image: <https://chem.libretexts.org/@go/page/172981>)

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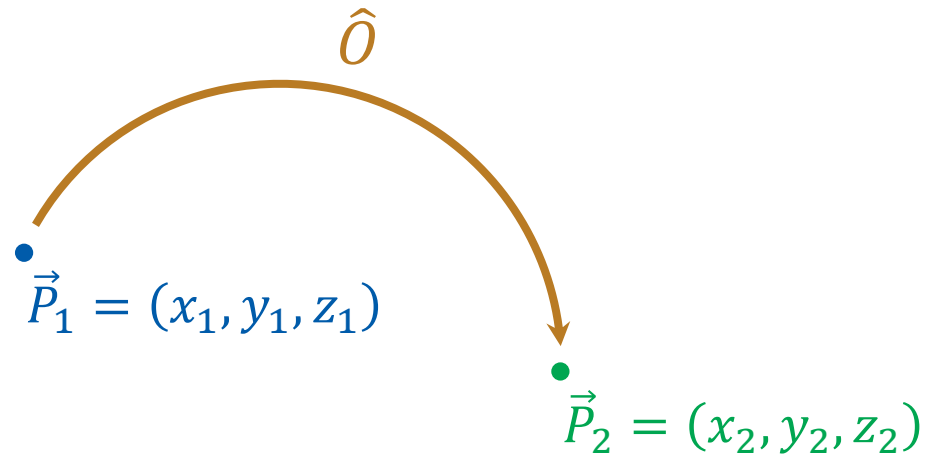
More Complex Molecules

How can we construct group orbitals of CH_4 ?



Group Theory!

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator

Inversion operator

Reflection operator

Rotation operator

Identity and Inversion Operators

- Identity operator $\hat{E}$: doing nothing

$$\hat{E}(x_1, y_1, z_1) = (x_1, y_1, z_1)$$

$$\hat{E}\vec{r}_1 = \vec{r}_1$$

- Inversion operator $\hat{i}$: inversion w.r.t. the origin

$$\hat{i}(x_1, y_1, z_1) = (-x_1, -y_1, -z_1)$$

$$\hat{i}\vec{r}_1 = -\vec{r}_1$$

Reflection Operators $\hat{\sigma}$

reflection through a plane

(The plane must be specified)

$$\hat{\sigma}_{(xy)}(x_1, y_1, z_1) = (x_1, y_1, -z_1)$$

$$\hat{\sigma}_{(xz)}(x_1, y_1, z_1) = (x_1, -y_1, z_1)$$

$$\hat{\sigma}_{(yz)}(x_1, y_1, z_1) = (-x_1, y_1, z_1)$$

Horizontal and Vertical Reflections

If the system has a principal axis,
we normally regard it as **the z axis**

- Horizontal reflection $\hat{\sigma}_h = \hat{\sigma}_{(xy)}$ (unique)
- Vertical reflection $\hat{\sigma}_v$: other reflections

$$\hat{\sigma}_{v(xz)}, \hat{\sigma}_{v(yz)}, \dots$$

Rotation Operators

- Proper rotation operator $\hat{C}_n$: counterclockwise rotation by $2\pi/n$
 - n applications of the rotation operator = 1 full rotation
 - The rotation axis must be specified
- Improper rotation operator $\hat{S}_n$: $\hat{C}_n$ followed by $\hat{\sigma}$
 - $\hat{\sigma}$: through a plane perpendicular to the rotation axis
 - The rotation axis must be specified

RS 13.1

Symbols for Rotational Symmetry



C_2 axis



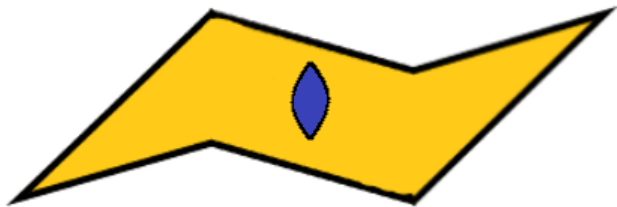
C_3 axis



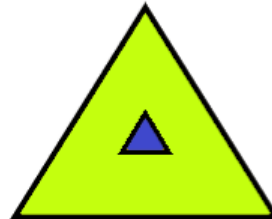
C_4 axis



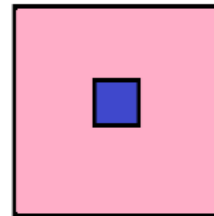
C_6 axis



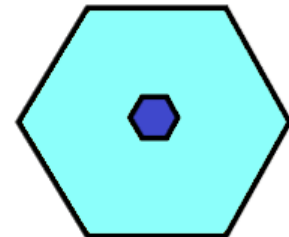
2-fold



3-fold



4-fold



6-fold

Symmetry Elements

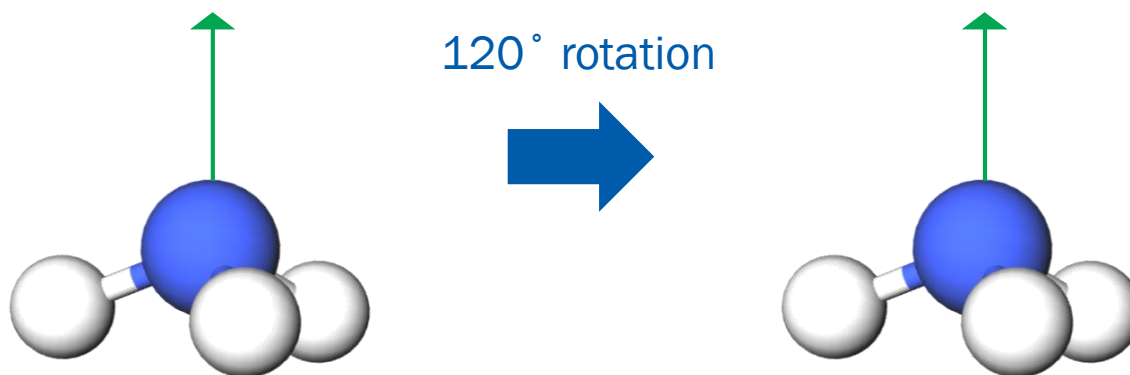
Point, line, or plane relative to which
the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol
- Inversion i : the origin
- Reflection σ : the reflection plane
- Rotation C_n, S_n : the rotation axis



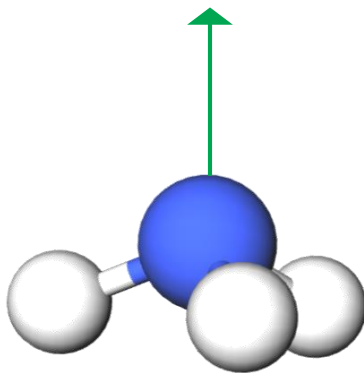
Symmetry of Molecule

If all atoms in the molecule are in the same conformation as before even after a symmetry operation, **the symmetry operator belongs to the molecule.**



Point Group

- A group formed by a set of point symmetry operators belonging to a molecule
- e.g. $\text{NH}_3 - \{\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c\}$

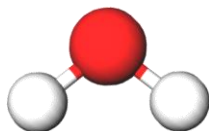
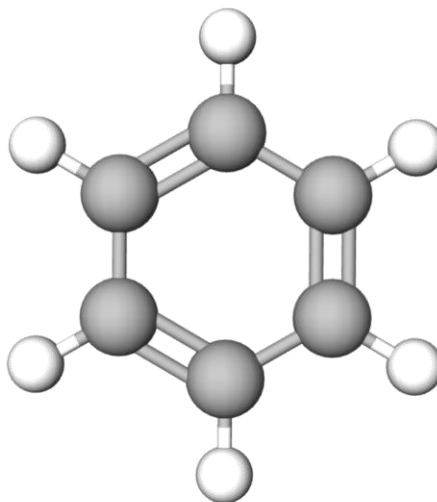
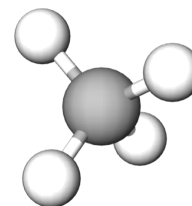
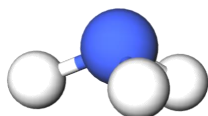
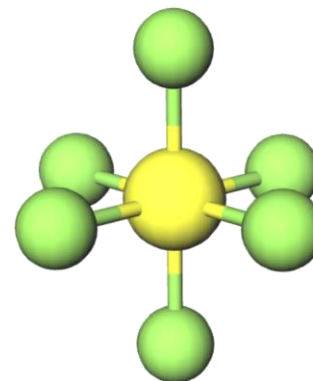
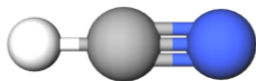
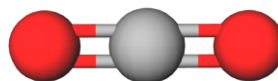


Schoenflies Symbols

Each point group has a unique symbol

Symbol	Symmetry elements
C_1	E
C_s	E, σ
C_i	E, i
C_n	E, C_n
S_n	E, S_n
C_{nv}	$E, C_n, n \times \sigma_v$
C_{nh}	E, C_n, σ_h
D_n	$E, C_n, n \times C_2$
D_{nh}	$E, C_n, n \times C_2, \sigma_h$
D_{nd}	$E, C_n, n \times C_2, n \times \sigma_v$
T_d	$E, 6 \times \sigma_d, 4 \times C_3, 3 \times S_4$
O_h	$E, 6 \times \sigma_d, 3 \times \sigma_h, 6 \times C_2, 4 \times C_3, 3 \times C_4, 3 \times S_4, 4 \times S_6$

Typical Point Groups

 C_{2v}  D_{6h}  T_d  C_{3v}  O_h  $C_{\infty v}$  $D_{\infty h}$

Character Table

lists the different symmetry types (“irreps”)
possible in a point group

- Character χ
 - shows how an object or mathematical function is affected by the corresponding symmetry operation of the group
 - 1: no change
 - -1: change of sign
 - otherwise: degenerate case (will be explained soon)

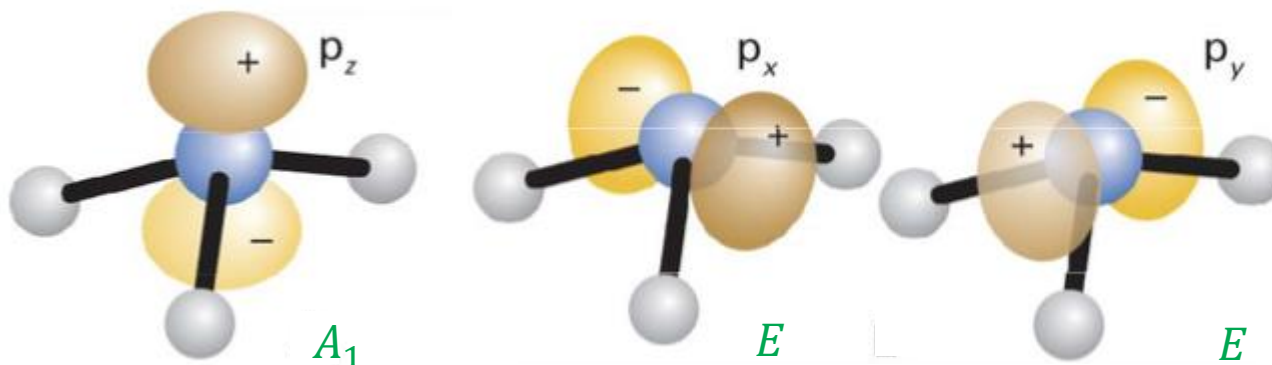
Case of C_{3v} Point Group

Class:	E	$2C_3$		$3\sigma_v$		
Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
A_1	1	1	1	1	1	1
A_2	1	1	1	-1	-1	-1
E	2	-1	-1	0	0	0

Class is constituted of symmetry operations
that are “physically identical”

Functions and Character Table

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



(Image: https://commons.wikimedia.org/wiki/File:Orbitals_of_NH3.png)

Character Table of C_{3v}

C_{3v}	E	$2C_3$	$3\sigma_v$	Basis components		
A_1	1	1	1	z		$x^2 + y^2, z^2$
A_2	1	1	-1		R_z	
E	2	-1	0	(x, y)	(R_x, R_y)	$(x^2 - y^2, xy)$ (xz, yz)

Not in text

Symmetry Operator and Overlap

- Overlap integral $S_{ij} = \int \psi_i^* \psi_j d\tau$
- Symmetry operation $\hat{R}$ can transform ψ_i and ψ_j

$$\hat{R}S_{ij} = \int (\hat{R}\psi_i^*)(\hat{R}\psi_j) d\tau$$

- Suppose that ψ_i^* and ψ_j are bases for the non-degenerate irrep Γ_a and Γ_b :

$$\hat{R}\psi_i^* = \chi_a(\hat{R})\psi_i^* \text{ and } \hat{R}\psi_j = \chi_b(\hat{R})\psi_j$$

Two Characters

$$\hat{R}S_{ij} = \int (\hat{R}\psi_i^*)(\hat{R}\psi_j) d\tau = \chi_a(\hat{R})\chi_b(\hat{R}) \int \psi_i^*\psi_j d\tau$$

For each $\hat{R}$ belonging to the molecule, we have $\hat{R}S_{ij} = S_{ij}$:

$$\chi_a(\hat{R})\chi_b(\hat{R}) \int \psi_i^*\psi_j d\tau = \int \psi_i^*\psi_j d\tau$$

For nonzero S_{ij} , we need

$$\chi_a(\hat{R})\chi_b(\hat{R}) = 1 \text{ (for all } \hat{R}\text{)}$$

Very Useful Result

For nonzero S_{ij} , we need

$$\chi_a(\hat{R})\chi_b(\hat{R}) = 1 \text{ (for all } \hat{R}\text{)}$$

Since Γ_a and Γ_b are non-degenerate irreps, $\chi = \pm 1$

Hence, for all $\hat{R}$,

$$\chi_a(\hat{R}) = \chi_b(\hat{R}) = 1 \text{ or } \chi_a(\hat{R}) = \chi_b(\hat{R}) = -1$$

ψ_i^* and ψ_j must belong to the same irrep

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Example: NH₃

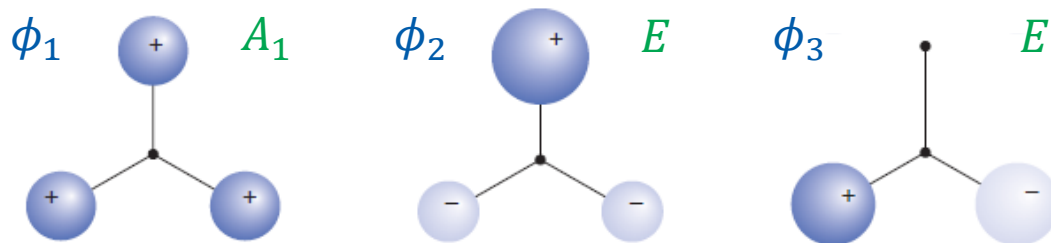
- Group orbitals of H atoms:

$$\phi_1 = \frac{1}{\sqrt{3}} (\psi_{A1s} + \psi_{B1s} + \psi_{C1s})$$

$$\phi_2 = \frac{1}{\sqrt{6}} (2\psi_{A1s} - \psi_{B1s} - \psi_{C1s})$$

$$\phi_3 = \frac{1}{\sqrt{2}} (\psi_{B1s} - \psi_{C1s})$$

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0



(Image: Atkins et al., *Shriver & Atkins' Inorganic Chemistry*, 5th Ed., Figure 6.20)

Not in text

Example: NH₃

Irrep	AOs of N atom	Group orbitals
A_1	ψ_{N2s}, ψ_{N2p_z}	ϕ_1
E	$\psi_{N2p_x}, \psi_{N2p_y}$	ϕ_2, ϕ_3

- Final MOs:

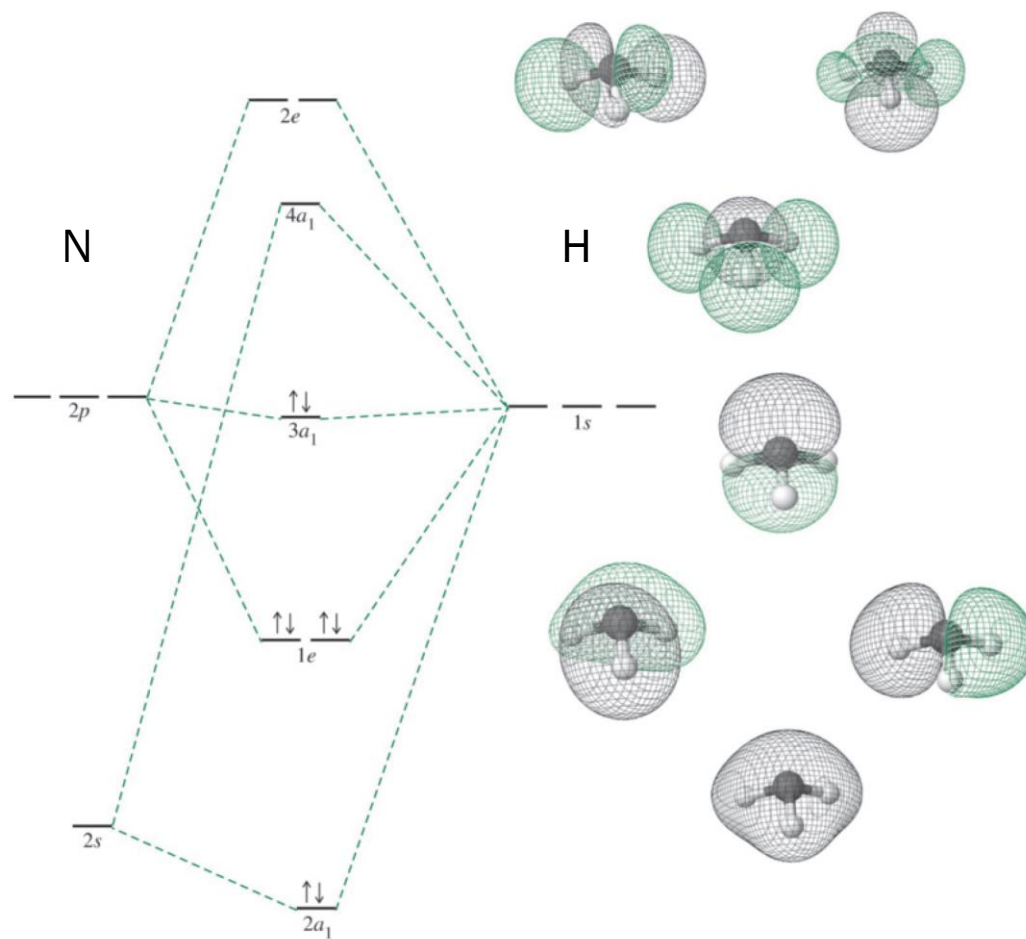
$$\phi_{a_1} = c_1\psi_{N2s} + c_2\psi_{N2p_z} + c_3\phi_1$$

$$\phi_e = c_4\psi_{N2p_x} + c_5\phi_2$$

$$\phi'_e = c_6\psi_{N2p_y} + c_7\phi_3$$

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MO Diagram of NH_3



(Image: Miessler et al., *Inorganic Chemistry*, 5th Ed., Figure 5.30)

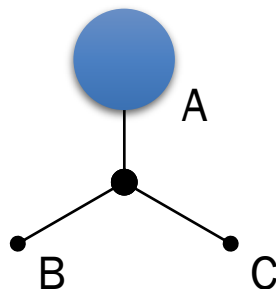
Projection Operator

$$\hat{P}_j = \sum_{\hat{R}} \chi_j(\hat{R}) \hat{R}$$

- Useful to find **group orbitals belonging to irrep j**
- The group orbitals are made of **symmetry-adapted linear combinations (SALCs)** of atomic orbitals

Finding SALCs for NH_3

- Pick one orbital, say ψ_{A1s} :



- Apply all operations in the point group: $\hat{E}$, $\hat{C}_3$, $\hat{C}_3^2$, $\hat{\sigma}_a$, $\hat{\sigma}_b$, $\hat{\sigma}_c$

$$\hat{E}\psi_{A1s} = \psi_{A1s}, \hat{C}_3\psi_{A1s} = \psi_{B1s}, \hat{C}_3^2\psi_{A1s} = \psi_{C1s}$$

$$\hat{\sigma}_a\psi_{A1s} = \psi_{A1s}, \hat{\sigma}_b\psi_{A1s} = \psi_{C1s}, \hat{\sigma}_c\psi_{A1s} = \psi_{B1s}$$

Finding SALCs for NH_3

$$\hat{E}\psi_{A1s} = \psi_{A1s}, \hat{C}_3\psi_{A1s} = \psi_{B1s}, \hat{C}_3^2\psi_{A1s} = \psi_{C1s}$$

$$\hat{\sigma}_a\psi_{A1s} = \psi_{A1s}, \hat{\sigma}_b\psi_{A1s} = \psi_{C1s}, \hat{\sigma}_c\psi_{A1s} = \psi_{B1s}$$

For each irrep, apply the projection operator:

$$\hat{P}_j\psi_{A1s} = \sum_{\hat{R}} \chi_j(\hat{R})\hat{R}\psi_{A1s}$$

e.g. for irrep A_1 ,

$$\hat{P}_{A_1}\psi_{A1s} = \psi_{A1s} + \psi_{B1s} + \psi_{C1s} + \psi_{A1s} + \psi_{C1s} + \psi_{B1s}$$

$$\therefore \phi_{A_1} = \frac{1}{\sqrt{3}}(\psi_{A1s} + \psi_{B1s} + \psi_{C1s})$$

Degenerate Case

- Different starting orbitals may give different group orbitals
- Combine them to obtain orthogonal orbitals
- e.g. irrep E in NH_3

$$\hat{P}_E \psi_{A1s} = 2\psi_{A1s} - \psi_{B1s} - \psi_{C1s} = \phi$$

$$\hat{P}_E \psi_{B1s} = 2\psi_{B1s} - \psi_{C1s} - \psi_{A1s} = \phi'$$

$$\hat{P}_E \psi_{C1s} = 2\psi_{C1s} - \psi_{A1s} - \psi_{B1s} = \phi''$$

$$\phi' - \phi'' = 3(\psi_{B1s} - \psi_{C1s}) \text{ is orthogonal to } \phi$$

$\therefore$ two group orbitals: ϕ and $(\phi' - \phi'')$